

# Functions: Composite, Inverse & Logarithmic

ANSWER KEY



## Domain and range

- Find the domain of  $f(x) = \frac{1}{x^2 - 9}$ .  
The denominator is zero when  $x = \pm 3$ . Domain: all real  $x$  except  $x = 3$  and  $x = -3$ .
- Find the domain of  $g(x) = \sqrt{7 - 2x}$ .  
The radicand must be non-negative:  $7 - 2x \geq 0 \Rightarrow x \leq \frac{7}{2}$ .
- Find the range of  $f(x) = -(x - 2)^2 + 5$ .  
Since  $-(x - 2)^2 \leq 0$  for all  $x$ ,  $f(x) \leq 5$ . Range:  $f(x) \leq 5$ .

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## Composite functions

- Given  $f(x) = x^2 + 1$  and  $g(x) = 2x - 3$ , find  $(f \circ g)(x)$ .  
 $f(g(x)) = (2x - 3)^2 + 1 = 4x^2 - 12x + 9 + 1 = 4x^2 - 12x + 10$ .
- Given  $f(x) = \sqrt{x + 4}$  and  $g(x) = x^2 - 4$ , find  $(f \circ g)(x)$ .  
 $f(g(x)) = \sqrt{(x^2 - 4) + 4} = \sqrt{x^2} = |x|$ .
- Using the same  $f$  and  $g$  as the previous question, find  $(g \circ f)(x)$  and state its domain.  
 $g(f(x)) = (\sqrt{x + 4})^2 - 4 = (x + 4) - 4 = x$ , valid for  $x \geq -4$  (the domain of  $f$ ).

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## Inverse functions

- Find  $f^{-1}(x)$  for  $f(x) = \frac{2x - 5}{3}$ .  
 $y = \frac{2x - 5}{3} \Rightarrow 3y = 2x - 5 \Rightarrow x = \frac{3y + 5}{2}$ . So  $f^{-1}(x) = \frac{3x + 5}{2}$ .
- Find  $f^{-1}(x)$  for  $f(x) = x^3 + 2$ .  
 $y = x^3 + 2 \Rightarrow x = \sqrt[3]{y - 2}$ . So  $f^{-1}(x) = \sqrt[3]{x - 2}$ .

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## Verifying inverses

- Verify that  $f(x) = 5x + 1$  and  $g(x) = \frac{x - 1}{5}$  are inverses by showing  $f(g(x)) = x$ .  
 $f(g(x)) = 5\left(\frac{x - 1}{5}\right) + 1 = (x - 1) + 1 = x$ . Confirmed.
- Does  $f(x) = x^2$  (for all real  $x$ ) have an inverse function? Explain.  
No.  $f$  is not one-to-one — it fails the horizontal line test, since for example  $f(2) = f(-2) = 4$ . An inverse function only exists if the domain is restricted (e.g. to  $x \geq 0$ ).

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## Logarithm laws

- 11 Expand  $\log_b \left( \frac{x^3 y}{z^2} \right)$  using the laws of logarithms.  
 $\log_b \left( \frac{x^3 y}{z^2} \right) = 3\log_b x + \log_b y - 2\log_b z.$
- 12 Condense  $2\log_5 x + 3\log_5 y - \log_5 z$  into a single logarithm.  
 $2\log_5 x + 3\log_5 y - \log_5 z = \log_5 \left( \frac{x^2 y^3}{z} \right).$
- 13 Evaluate  $\log_2 32 - \log_2 4$  without a calculator.  
 $\log_2 \left( \frac{32}{4} \right) = \log_2 8 = 3.$
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## Solving logarithmic equations

- 14 Solve  $\log_4 (x^2) = 3$ .  
 $x^2 = 4^3 = 64 \Rightarrow x = \pm 8$ . Both satisfy the domain requirement  $x^2 > 0$ . Solutions:  $x = 8$  or  $x = -8$ .
- 15 Solve  $\ln x + \ln(x - 3) = \ln 10$ .  
 $\ln [x(x - 3)] = \ln 10 \Rightarrow x^2 - 3x - 10 = 0 \Rightarrow (x - 5)(x + 2) = 0 \Rightarrow x = 5$  or  $x = -2$ . Since  $\ln(x - 3)$  requires  $x > 3$ , reject  $x = -2$ . Solution:  $x = 5$ .
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## Exponential equations and logs as inverses

- 16 Solve  $e^{2x-1} = 20$ , giving your answer to three decimal places.  
 $2x - 1 = \ln 20 \approx 2.996 \Rightarrow x = \frac{1 + 2.996}{2} \approx 1.998.$
- 17 The functions  $f(x) = \log_3 x$  and  $g(x) = 3^x$  are inverses. Use this fact to solve  $3^x = 81$  without a calculator, and state the value of  $g(f(16))$ .  
 $81 = 3^4$ , so  $x = 4$ . Since  $f$  and  $g$  are inverses,  $g(f(16)) = 16$ .
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## Transformations of functions

- 18 The graph shows  $f(x) = x^2$  (teal) and  $g(x) = (x - 2)^2 + 1$  (indigo). Describe the transformation that maps  $f$  onto  $g$ .  
 $g$  is  $f$  shifted right 2 units and up 1 unit — a horizontal translation of +2 followed by a vertical translation of +1.
- 19 If  $h(x) = -f(x + 3)$  where  $f(x) = x^2$ , describe the transformations applied to  $f$  to obtain  $h$ .  
Shift  $f$  left 3 units (from  $x + 3$ ), then reflect the result over the  $x$ -axis (from the negative sign outside).

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## A well-designed word problem: investment growth and doubling time

- 20** This question has three parts. An investment fund grows according to  $A(t) = A_0 e^{0.045t}$ , where  $A_0$  is the initial amount (in dollars) and  $t$  is time in years. (a) If  $A_0 = 8000$  dollars, find  $A(10)$  to the nearest dollar. (b) Find the time  $t$ , to the nearest tenth of a year, for the investment to double. (c) Solve for  $t$  in terms of  $A$  and  $A_0$  — that is, find the inverse function  $t(A)$ .

(a)  $A(10) = 8000e^{0.45} \approx 8000(1.5683) \approx 12546$  dollars. (b) Doubling requires  $e^{0.045t} = 2 \Rightarrow 0.045t = \ln 2 \Rightarrow t = \frac{\ln 2}{0.045} \approx 15.4$  years. (c) From  $A = A_0 e^{rt}$ :  
 $\frac{A}{A_0} = e^{rt} \Rightarrow \ln\left(\frac{A}{A_0}\right) = rt \Rightarrow t(A) = \frac{1}{0.045} \ln\left(\frac{A}{A_0}\right).$